Project Specification

Background

Cosmetic automobile modifications have become a booming business. The custom wheel rim has become a popular modification for car enthusiasts. However, to this date, rims have not been enhanced with lights. This project presents a design of a rim-mounted LED display. The system will consist of a controllable array of LEDs, which will create a visual pattern when spinning. Using an easy-to-use LCD display, the user will select from various configurations which will determine the output of the display. A vehicle mounted controller will wirelessly communicate with a microcontroller that will drive the LED display. The illuminated wheel display will provide an aesthetic advantage over standard, non-illuminated custom wheel rims.

The system will be sold at specialty automobile component shops, including car audio and custom modification specialists.

Users

	Background	**Needs**
Car Enthusiast	Always looking for the newest accessories to improve their car's aesthetics Likely has some experience in car maintenance and modifications	To have an innovative unique automobile appearance. To quickly learn the controls and features of their Wheel Blazers
Car Accessory Sales Person	Experienced in selling car accessories and vehicle modifications. New flashy car aesthetics will be easy to install and attract customers	Easily explained user interface which can be stylishly integrated into the vehicle dash

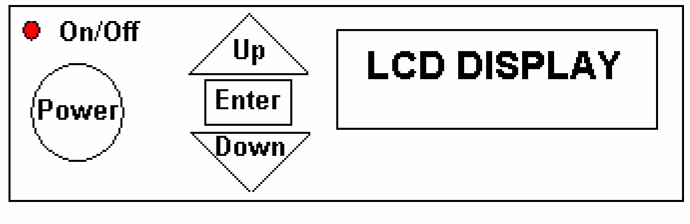


Figure 1 – Dashboard Mounted Control Unit Faceplate

Figure 2 – LED Display mounted on rim. All electronic components are hidden from view.

Functional Requirements

Priority	Name	Measure	Description
Must	Simple on/off function of the control unit	By design	No signal should be transmitted when the unit is off
Must	Present mode identification on control unit	By design	Feedback must be provided to user to indicate the present mode of operation
Must	Quick and simple mode	By design	Control unit allows for

	control		easy switching of the display mode to minimize distractions from driving
Should	Obtain audio signal from an external source	By design	Analog audio signal to be used to synchronize the visual effects with the music
Must	Transmit mode information to the LED display unit	5 seconds	Mode is received by LED display unit and the visual effect is changed accordingly
Must	Wireless transmission range	~10 feet	Control unit must be able to communicate with the LED display unit located on the rear wheel of a large passenger vehicle
Must	Control unit power supply	12 V	Control unit operates off the standard 12 V vehicle power system
Must	Implement an array of LEDs	By design	The LED display unit must contain an array of LEDs that can be independently controlled to produce various visual effects
Must	Wheel mounted LED display unit power source	4 V - 10 V	Power source must be suitable to power the controller, the wireless receiver and the LED array
Should	Implement a mechanical to electrical energy conversion mechanism	By design	Increase operating time by harnessing available mechanical energy
Must	Decode signal from control unit	By design	LED controller must decode the transmitted signal to determine the appropriate mode
Must	Drive LED array	By design	Based on the current mode, the LED controller must toggle LEDs accordingly

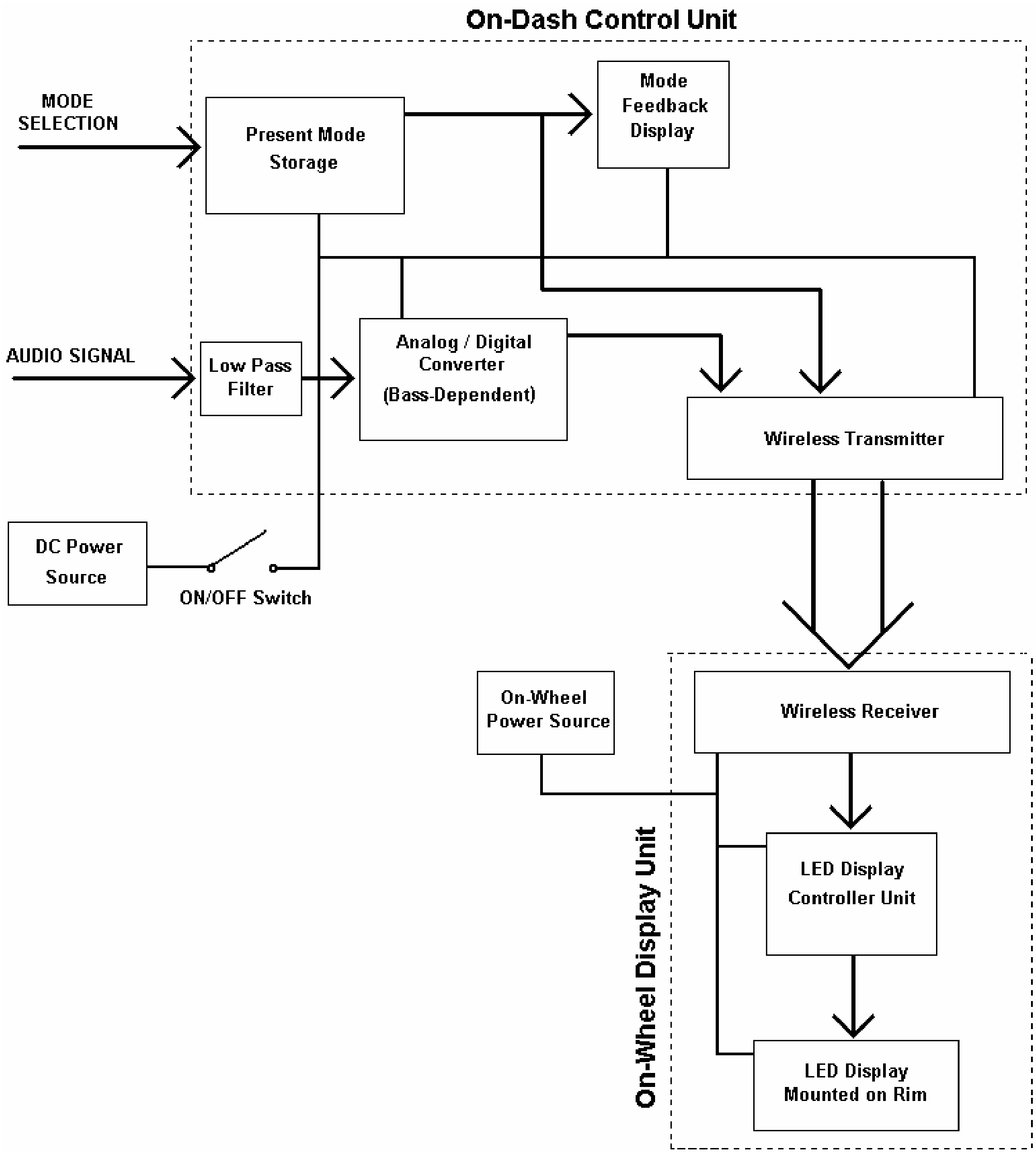
Non-Functional Requirements

Priority	Factor	Description
Must	Controller Position	The control unit must be mounted on the dash of the vehicle
Must	Controller Size	The control unit must not be larger than 20 cm by 10 cm by 10 cm
Must	Controller Weight	The control unit must weigh less than 2 kg
Should	Controller Current Draw	The control unit should draw no more than 300 mA of current
Should	Controller Operating Environment	The controller is expect to operate in the range of 0 degrees Celsius to 40 degrees Celsius
Must	LED Display Unit Position	The LED display unit must be securely mounted on the rim/rim-attachment
Must	LED Display Unit Size	The display unit must be small enough to be mounted on a rim
Must	LED Display Unit Weight	The display unit's weight must not severely impede the rotation of the rim
Should	LED Display Unit Operating Environment	The LED display unit should be protected from moisture. It should be able to operate at a temperature between 0 degrees Celsius and 40 degrees Celsius. It must be able to withstand motion and vibration without interrupting operation.

Cost

	Prototype
Quantity	1
Estimate	~$1100

Block Diagram



Risks

Level	Description	Response
High	The hardware required to be attached to the rim will prove too large and heavy. This could result in a disruption in the spinner rim's operation or may simply demonstrate that the system is	Initially, different models of spinner rims will be investigated in hopes of finding a model with sufficient clearance for the required hardware. If this still proves impossible, the design

	unfeasible for the selected rims.	will be changed such that it is implemented on a standard automotive hubcap.
Medium	The power required by the LEDs limits the use of the Wheel Blazers to an unacceptably short life.	Attempt to recharge the battery while the vehicle is in motion. This could perhaps be accomplished by harnessing the wheel's rotational energy, increasing the number of batteries, or implementing a system of brushes to transmit electricity to the moving rim.
Low	Interference in the transmitted signal will cause the on-wheel system to receive erroneous commands.	Investigate changing the location of the transmitter and receiver or using a different system to relay the information. This could be as easy as using a system that transmits on a different frequency or that uses a different communication scheme.
Low	An appropriate rim cannot be purchased or shipped on time to be used for the prototype demonstration.	Implement the spinning LED display on a standard, easily obtained vehicle hubcap.

Timeline Plan

	May				June				July					Hours				Total
Tasks	1	2	3	4	5	6	7	8	9	10	11	12	13	ZB	CM	JN	MS	
Define Hardware Requirements	+													12	12	12	12	48
Verify Block Diagram		+												5	5	5	5	20
Block Verification		d												5	5	5	5	20
Design LED Grid Layout			+												2	2		4

Design LED Control Circuit			+											15			15	30
Breadboard LED Control Circuit			+											5			5	10
Design Bass Signal Conversion			+												10	10		20
Design Controller Input Circuit				+										2	10	10		22
Design Communication System				+										10			10	20
Review Detailed Design					+									5	5	5	5	20
Detailed Design					d									5	5	5	5	20
Build Circuitry for Controller						+		+							15	15		30
Midterm Break							---											0
Build Circuitry for LED Display						+		+						15			15	30
Assemble LED Display onto Rim									+	+					10	10		20
Prepared Testing Checklist										+					5	5		10
Testing and Debugging										+	+			10	5	5	10	30
Prototype Testing Checklist											d				5	5		10
Final Debugging												+		10	5	5	12	32
Prototype Demonstration												d		2	2	2	2	8
Experience Report													d	3	3	3	3	12
Begin Draft of Final Report													+	2	2	2	2	8
Total Hours														**106**	**106**	**106**	**106**	**424**

Legend

+	Active Time	ZB	Zac Balson
d	Deliverable	CM	Carlos Midence
		JN	Jonathan Neill
		MS	Matt Strickland

Budget

Item	Description	Part No.	Manufacturer	Source	Price
Controller	Basic Stamp 2	BS2SX-IC	Parallax	ECE Stores	73.42
Programming Board	Basic Stamp 2 Starter Kit		Parallax	Digikey.ca	207.56
Wireless Transmitter	RF Transmitter 433MHZ SMT	TXM-433-KH	Linx Technologies Inc	Digikey.ca	13.03
LCD Screeen	Backlit 2 line, 16 Char. Display	N/A	Matrixorbital	Matrixorbital.com	90.00
A/D Converter	IC 8-BIT SERIAL CONTROL A/D 8-D	TLC549IP	Texas Instruments	Digikey.ca	3.24
Rim	Chrome/Alloy Rim	N/A			200.00
Wireless Receiver	RF Receiver 433MHZ Surface Mount	RXD-433-KH	Linx Technologies Inc	Digikey.ca	19.82
Display Controller	Basic Stamp 2	BS2SX-IC	Parallax	ECE Stores	73.42
LED Array	5mm super-bright blue LED	55-558SB-10		Active Electronics	110.00
Enclosure	Plastic Case			www.polycase.com	10.00
Power Supply	Power Supply 12V Alimentation			ECE Stores	
Circuit Board Printing				UW facilities	50.00
Poster					200.00
Misc.	Buttons, wires, resistors, capacitors, soldering				50.00

Need	Qty	@	Total	Cash			In-kind
				SPF	Team	E&CE	Team
BASIC Stamp	2	73.42	146.84			146.84	
BASIC Stamp 2 Starter Kit	1	207.56	207.56	207.6			
Wireless TRX	1	13.03	13.03	13.03			
Wireless RXR	1	19.82	19.82	19.82			
AD Converter	1	3.24	3.24	3.24			
Rim	1	200	200	200			
LED/10	2	54.99	109.98	110			
LCD Screen	1	90	90	90			
Enclosure	1	10	10	10			
Power Supply	1	0	0	0		0	
Circuit Boards	1	50	50	50			
Poster	1	200	200		200		
Misc.	1	50	50	50			